Bacterial pathogenesis

Chapter 7

Symbiosis

- **Symbiosis:** "live together“

- Two organisms live in an association with one another

- Three types of relationships based on the quality of the relationship for each member of the symbiotic association

Mutualism

- **Both members of the association benefit**

- Classic mutualistic association: **lactic acid bacteria** that live on the vaginal epithelium of a woman

Woman: provided habitat with a constant temperature and supply of nutrients (glycogen) in exchange for the production of lactic acid

Bacteria: protects the vagina from colonization and disease caused by yeast and other potentially harmful microbes.

Commensalisms

- There is no apparent benefit or harm to either member of the association
- *Staphylococcus epidermidis*, a consistent inhabitant of the skin of humans

- The bacterium produces lactic acid that protects the skin from colonization by harmful microbes

- It has been suggested that other metabolites that are produced by the bacteria are an important cause of body odors and possibly associated with certain skin cancers

What is a Parasite?

- An organism that grows, feeds, and lives on or in a host, benefiting from the host while providing no benefit to its survival.

Parasitism & Disease

- The parasite causes **harmful effects** on the host and may cause host damage.

- When host damage is sufficient to cause disease, the parasite is considered **pathogenic**.

- Some parasitic bacteria can exist as normal flora and cause disease when an **opportunity arises**.

PRINCIPLES OF PATHOGENESIS

- **Pathogen:** A microorganism that can cause disease.

- **Opportunistic pathogens:** Microorganisms that rarely cause disease in healthy individuals but can cause serious infections in immunocompromised patients.

Examples include normal flora and some parasites.

PATHOGENICITY AND VIRULENCE

- **Pathogenicity:** The ability of a microorganism to cause disease in a host.

- **Virulence:** The degree or severity of pathogenicity of a microorganism; it is a quantitative measure of pathogenicity.

  - It can be assessed by the **number of microorganisms required to cause disease.**

INFECTIOUS DOSE & VIRULENCE (LD_{50} and ID_{50})

- LD_{50} (50% lethal dose): number of organisms required to kill 50% of exposed hosts.

- ID_{50} (50% infectious dose): number of organisms required to cause infection in 50% of exposed hosts.

Lower LD_{50} or ID_{50} means greater virulence, because fewer organisms are needed to cause death or disease.

TYPES OF PATHOGENS

A - Potential Pathogens (Opportunistic Pathogens)

Normal bacterial flora that can cause disease under certain conditions:

- *Staphylococcus aureus*
- *Streptococcus pneumoniae*
- *Haemophilus influenzae*

Normally live in a commensal relationship with the host without causing disease. They can cause disease when an opportunity arises due to a compromise or weakness in the host, such as:

Anatomical barriers, Tissue resistance, Immune defenses

- Opportunistic infection occurs when host defences are weakened or disrupted.

TYPES OF PATHOGENS

B. Obligate Pathogens

- Normally do not live in association with the host as part of the normal flora.

- Their presence in the host is generally associated with disease.

- Often have mechanisms that allow them to survive and resist elimination by the host's defenses.

INTRACELLULAR AND FACULTATIVE BACTERIAL PATHOGENS

- Obligate intracellular pathogens:

- Some bacterial pathogens can grow and reproduce only inside host cells.

Examples: *Chlamydia* and *Rickettsia*

- Facultative intracellular pathogens:

Many bacteria can grow both inside and outside host cells, and can also be cultured on bacteriologic media.

Examples:

Listeria monocytogenes*, *Salmonella enterica*, and *Mycobacterium tuberculosis

Why Does Infectious Dose Vary?

The infectious dose differs greatly among pathogenic bacteria and depends on virulence factors that include:

- Adhesins/pili (attachment to mucous membranes)
- Exotoxins & endotoxin (tissue damage and disease)
- Capsules (evasion of host defenses)
- Resistance to nonspecific defenses (e.g., survival of stomach acid)

WHY DO PEOPLE GET INFECTIOUS DISEASES?

- Infection occurs when microorganisms overcome the host's defences

The microorganism or its products will cause tissue damage and symptoms such as fever and inflammation.

- Two major factors influence whether infection occurs:
  - Infectious dose.
  - Pathogen virulence.
  - ??

Virulence factors determine the type and severity of disease.
Example: *E. coli,* different strains produce different exotoxins, leading to different clinical outcomes:

- Watery (nonbloody) diarrhea : one exotoxin
- Bloody diarrhea is another exotoxin

DETERMINANTS OF VIRULENCE

Virulence determinants are the genetic, biochemical, and structural features of a pathogen that enable it to cause disease in a host.

1- Genetic features: (Pathogenicity genes carried on a plasmid)

Example: virulence genes in Salmonella

2- Biochemical features (Toxins and enzymes that damage host tissues

Example: botulinum toxin

3- Structural features (Capsules, pili, or surface proteins that promote colonization or immune evasion)

Example: capsule of Streptococcus pneumoniae

DETERMINANTS OF VIRULENCE

1- Single Determinant of Virulence: Cause disease through one major virulence factor, (toxin).

- *Clostridium tetani* (Tetanus toxin affect muscle spasms
- *Corynebacterium diphtheriae* (Diphtheria toxin casue tissue damage)

2- Multiple Determinants of Virulence: Pathogens possess multiple virulence factors, that are cause different diseases in different tissues.

- *Staphylococcus aureus* (causes skin, soft-tissue, bloodstream, and other infections)
- *Streptococcus pyogenes* (causes pharyngitis, skin infections, and invasive disease)
- *Pseudomonas aeruginosa* (causes infections in different tissues in case of immunodeficiency.

ASYMPTOMATIC (SUBCLINICAL) INFECTION

- **Healthy immune** response means better protection against infection and disease.

- **Asymptomatic infection:** The host acquires the microorganism, but no clinical disease develops because the immune defences successfully control the infection.

Although the patient has no symptoms, **evidence of infection** may still be detected, such as:

- **Specific antibodies** against the microorganism.
- **Detection or isolation of the microorganism**

Bacteria cause disease by two major mechanisms

1. TOXIN PRODUCTION
2. INVASION & INFLAMMATION

1. TOXIN PRODUCTION

- **Exotoxins:** protein (polypeptide) toxins that are produced and released by bacteria.

- **Endotoxin:** lipopolysaccharide (LPS), an integral component of the outer membrane of Gram-negative bacteria.

- Both can produce systemic symptoms such as fever, inflammation, and shock.
- Toxins may cause disease even when bacteria are no longer present.

Bacteria cause disease by two major mechanisms

1. TOXIN PRODUCTION
2. INVASION & INFLAMMATION

2. INVASION & INFLAMMATION

- Invasive bacteria enter and multiply in host tissues.

- Their presence triggers an inflammatory response that casue erythema, edema, warmth, and pain.

- Tissue damage may result from both bacterial activity and the host inflammatory response.

COMMUNICABLE vs. NONCOMMUNICABLE INFECTIONS

1 - Communicable infections

Many, but not all, infections are communicable (they can be spread/transmitted from one host to another).

Example: Tuberculosis (TB)

Caused by *Mycobacterium tuberculosis*

Spread from person-to-person through airborne particles. released during coughing, speaking, or breathing.

COMMUNICABLE vs. NONCOMMUNICABLE INFECTIONS

2- Noncommunicable infections

- Not transmitted directly from one host to another.

Example: Botulism

- Caused by *Clostridium botulinum* toxin.

- Usually acquired by consuming contaminated food containing preformed toxin.
- The illness is not spread person to person.

Contagious Infection means:

A highly communicable infection that spreads easily from person to person.

PATTERNS AND STATES OF INFECTION

Epidemiological Patterns

- **Epidemic infection:** Occurs more frequently than expected in a specific population or geographic area.

- **Pandemic infection:** An epidemic that has spread across multiple countries or continents, with worldwide or very widespread distribution.

- **Endemic infection:** An infection that is constantly present at a relatively stable level in a specific population or geographic area.

Clinical States of Infection

Inapparent (Subclinical) infection: An infection that produces no obvious symptoms and may be detected only by:

- A rise in antibody titer, or
- Isolation/detection of the organism.

Latent infection: A state in which the pathogen remains in the host in a dormant or inactive form and may reactivate later, causing recurrence of symptoms.

Example: Herpes simplex virus

Chronic infection: An infection that persists for weeks, months, or longer, with the pathogen continuing to be present and potentially produced or shed by the host.

Chronic carrier state: The pathogen persists and may continue to be shed, with or without symptoms.

Example: *Salmonella Typhi* carrier state ("**Typhoid Mary**")

Chronic carriers (important **source of infection**)

"TYPHOID MARY"

MOST HARMF



The Extraordinary Predicament of Mary Mallon, a Prisoner on New York's Quarantine Hospita

IT is probable that Mary Mallon is a prisoner for life—and yet she has committed no crime, has never been accused of an immoral or wicked act, and has never been a prisoner in any court, nor has she been sentenced to imprisonment by any judge.

Mary Mallon is a cook by profession. She has served in the kitchens of many New York millionaires with entire satisfaction for many years.

Mary Mallon for more than two years has been a prisoner on New York's quarantine island, along with the unfortunates who are from time to time removed to this isolated spot because they are suffering from smallpox, scarlet fever or other contagious diseases.

But while Mary sees these unfortunate victims of such dire diseases come on the hospital boat and, in due time, return to their homes and friends—Mary stays on forever.

There is probably in the whole wide world no prisoner that can furnish a parallel to the extreme misfortune which has brought Mary Mallon to North Brother Island. Through no fault of hers, Mary Mallon is a living, walking incubator of typhoid fever germs. Every day for two years the officials of the New York Board of Health have examined Mary, and they have been discouraged to find a bountiful supply of new typhoid fever bacilli freshly made each twenty-four hours by Mary.

Mary Mallon, in the five years before authorities got their hands on her, was responsible for a number of twenty cases of typhoid fever, some of which proved fatal. So far as is known, the victims had the disease, and in one case only did. But somewhere in her anatomy, perhaps in the gall duct, there is a never failing supply of typhoid fever germs. To the positive knowledge of the Board of Health Mary has been a typhoid bacilli for seven years. They are also in some her kidneys.

By Dr. Wm. H. Park, New York Board of Health.

MARY MALLON is the chief of fifty persons, dozens of whom have been discovered within the past two years, at the city institution, which we know as typhoid bacilli carriers. She came in the most unexpected way, where we are accustomed, because of the number of persons in whom she has disseminated the disease.

The study of the case leads us to believe that the typhoid germs lodge in the gall bladder, where they are undisturbed. When these germs are carried through the body by the bile,

Also is confined on North Brother Island, where she has been a prisoner for two years, and where, of course, she will remain indefinitely, she is a large, healthy looking woman, a cheerful soul, and there is nothing to her outward appearance to indicate that she is other than normal. She is, of course, segregated with the typhoid patients. When we remember that she has been spreading the contagion for many years, it is clear that she will be a prisoner on North Brother Island for a long time, perhaps for the remainder of her life, unless she can be convinced that the typhoid germs and bacilli undermine have been eliminated from her body.

Every effort has been made by the health authorities to cure the unfortunate woman, but so far without success. Examinations in search of the typhoid germs and bacilli in the various excretions we have cited have one and all failed to show the presence of typhoid. Perhaps we have only so far been unable to detect the germs, or perhaps at the present moment science is making some wonderful discovery in medical science which enables to make a cure in this. It is satisfactory enough for the woman, but in the main duty of the health authorities to safeguard the public from such a menace.

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The Official R

IN the Winter of 1906 I was called on to investigate a household epidemic of typhoid fever which had broken out in the better part of that suburb at Oyster Bay, N. Y. The epidemic had been studied carefully immediately after it took place, but its cause had not been ascertained with any degree of certainty. An avowed discussion on the subject of the property.

The essential facts concerning the investigation follow:

On Oyster Bay, in the Summer of 1906, six persons in a household of seven were attacked with typhoid fever. The place was large, surrounded with ample grounds, in a pleasant part of the village, and had been rented for the Summer by a New York banker.

The first person was taken sick on August 27, and the last on September 3. The diagnosis of typhoid was positive. Two of the patients were sent to the Nassau Hospital at Mineola. The others were attended by resident physicians of Oyster Bay. None of the subsequent cases apparently resulted from the first, although the limited time the first to the last might permit of this assumption. But whether the disease was communicated from one person to another after the first case occurred was not a matter of great consequence. The most important question was how the first case originated.

Germs Source a Mystery.

Typhoid fever is an enteric disease in direct line, according to the views prevalent with the medical profession here, at the time of the outbreak as other case was known. These deliriums.

The milk supply of the house was the same as used by each of the other persons in the village, all of whom remained well. The cream also was from a source which supplied several other families in the vicinity.

To the first investigators it seemed that the water must have been contaminated. They were

STAGES OF BACTERIAL PATHOGENESIS

A generalized sequence of the stages of infection is as follows:

1- Transmission from an external source into the portal of entry.

2- Evasion of primary host defenses such as skin or stomach acid.

3- Adherence to mucous membranes, (pili, fimbriae, or adhesins).

4- Colonization by growth of the bacteria at the site of adherence.

5- Disease symptoms caused by toxin or invasion accompanied by inflammation.

6- Host responses, both nonspecific and specific (immunity).

7- Progression, persistence, or resolution of the infection and disease.

1. Transmission:

It is important to understand of the mode of transmission of bacteria and other infectious agents in order to know how to interrupt the chain of transmission and then prevent infectious diseases.

Different modes of transmission of infectious diseases:

1- Human-to-human.

2- Transmition from nonhuman sources: soil, water, and animals.

3- Fomites (towels, clothes, medical equipment)

TABLE 7–1 Important Modes of Transmission

Mode of Transmission	Clinical Example	Comment
I. Human to human		
A. Direct contact	Gonorrhea	Intimate contact (e.g., sexual or passage through birth canal)
B. No direct contact	Dysentery	Fecal–oral (e.g., excreted in human feces, then ingested in food or water)
C. Transplacental	Congenital syphilis	Bacteria cross the placenta and infect the fetus
D. Blood-borne	Syphilis	Transfused blood or intravenous drug use can transmit bacteria and viruses; screening of blood for transfusions has greatly reduced this risk
II. Nonhuman to human		
A. Soil source	Tetanus	Spores in soil enter wound in skin
B. Water source	Legionnaire's disease	Bacteria in water aerosol are inhaled into lungs
C. Animal source		
1. Directly	Cat-scratch fever	Bacteria enter in cat scratch
2. Via insect vector	Lyme disease	Bacteria enter in tick bite
3. Via animal excreta	Hemolytic-uremic syndrome caused by *E. coli* O-157	Bacteria in cattle feces are ingested in undercooked hamburger
D. Fomite source	Staphylococcal skin infection	Bacteria on an object (e.g., a towel) are transferred onto the skin

Shiga toxin-producing E. coli (STEC) can be found in ground meat, raw milk, and fecal contamination of vegetables

Vertical transmission: From mother to offspring, that occurs in the following routes:

1- Cross placenta,
2- Birth canal during birth,
3- Via breast milk.

Horizontal transmission: Describing transmission from person-to-person.

VERTICAL AND HORIZONTAL TRANSMISSION

1- Vertical Transmission: Transmission of an infectious agent from a mother to her offspring.

1. **Transplacental transmission** (across the placenta). *Examples:* Treponema pallidum (syphilis), Toxoplasma gondii, and Cytomegalovirus (CMV).

2. **During childbirth** (birth canal). *Examples:* Neisseria gonorrhoeae, Chlamydia trachomatis, Herpes simplex virus, Group B strep. (GBS)).

3. **Via breast milk:** *Examples:* HIV and Human T-cell leukemia virus type 1 (HTLV-1).

2- Horizontal Transmission: Transmission of an infectious agent from one person to another.

Examples: Many:

Influenza virus (respiratory droplets/aerosols)
Mycobacterium tuberculosis (airborne transmission)
Salmonella (fecal/oral transmission)
HIV (sexual contact or blood exposure)

ROUTES OF TRANSMISSION

- Pathogens leave the infected host through the respiratory, gastrointestinal, genitourinary, and other routes.

- Transmission may occur through respiratory droplets / aerosols or fecal contamination of food and water.

- Other routes include sexual contact, urine, direct skin contact, blood transfusion (*Screening blood donors for infectious pathogens*), and contaminated needles.

You are expected to know all of these infectious agents by the third year of your studies.

TABLE 7–4 Transmission of Important Waterborne Diseases

Portal of Entry	Pathogen	Type of Organism[1]	Disease
Gastrointestinal tract			
1. Ingestion of drinking water	*Salmonella* species	B	Diarrhea
	Shigella species	B	Diarrhea
	Campylobacter jejuni	B	Diarrhea
	Norovirus[2]	V	Diarrhea
	Giardia lamblia	P	Diarrhea
	Cryptosporidium parvum	P	Diarrhea
2. Ingestion of water while swimming[3]	*Leptospira interrogans*	B	Leptospirosis
Respiratory tract			
Inhalation of water aerosol	*Legionella pneumophila*	B	Pneumonia (Legionnaire's disease)
Skin			
Penetration through skin	*Pseudomonas aeruginosa*	B	Hot-tub folliculitis
	Schistosoma mansoni	H	Schistosomiasis
Nose			
Penetration through cribriform plate into meninges and brain	*Naegleria fowleri*	P	Meningoencephalitis

[1] B, bacterium; V, virus; P, protozoa; H, helminth.

[2] Formerly called Norwalk-like viruses.

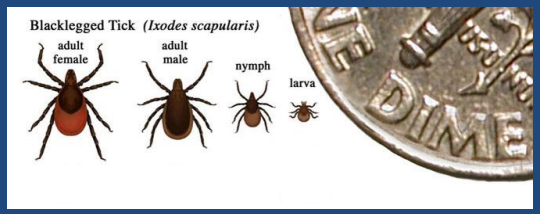
[3] All of the organisms that cause diarrhea by ingestion of drinking water also cause diarrhea by ingestion of water while swimming.

Vector-borne transmission: occurs through the bites of infected insects, such as mosquitoes and ticks.

Examples: Lyme disease (Borrelia), Rocky Mountain spotted fever (*Rickettsia)*, ehrlichiosis, relapsing fever, and tularemia.



Lyme disease is caused by the bacterium *Borrelia **burgdorferi*** and **is trammitted** to humans through the bite of infected **ticks.** Typical include fever, headache, fatigue, and a characteristic skin rash called **erythema migrans**



- Some diseases are transmitted by insects (vector)

TABLE 7–6 Bacterial Diseases Transmitted by Insects

Bacterium	Insect	Reservoir	Disease
Gram-negative rods			
Yersinia pestis	Rat fleas	Rodents (e.g., rats and prairie dogs)	Plague
Francisella tularensis	Ticks (*Dermacentor*)	Many animals (e.g., rabbits)	Tularemia
Spirochetes			
Borrelia burgdorferi	Ticks (*Ixodes*)	Mice	Lyme disease
Borrelia recurrentis	Lice	Humans	Relapsing fever
Rickettsias			
Rickettsia rickettsii	Ticks (*Dermacentor*)	Dogs, rodents, and ticks (*Dermacentor*)	Rocky Mountain spotted fever
Rickettsia prowazekii	Lice	Humans	Epidemic typhus
Ehrlichia chaffeensis	Ticks (*Dermacentor, Ixodes*)	Dogs	Ehrlichiosis

You are expected to know all of these infectious agents by the third year of your studies.

Important portals of entry

- Respiratory tract.
- Gastrointestinal tract.
- Genital tract.
- Skin

- **Zoonotic disease:** Transmission of a pathogen between animals and human being.

TABLE 7–3 Portals of Entry of Some Common Pathogens

Obligatory!/ know all species

Portal of Entry	Pathogen	Type of Organism[1]	Disease
Respiratory tract	*Streptococcus pneumoniae*	B	Pneumonia
	Neisseria meningitides	B	Meningitis
	Haemophilus influenzae	B	Meningitis
	Mycobacterium tuberculosis	B	Tuberculosis
	Influenza virus	V	Influenza
	Rhinovirus	V	Common cold
	Epstein–Barr virus	V	Infectious mononucleosis
	Coccidioides immitis	F	Coccidioidomycosis
	Histoplasma capsulatum	F	Histoplasmosis
Gastrointestinal tract	*Shigella dysenteriae*	B	Dysentery
	Salmonella typhi	B	Typhoid fever
	Vibrio cholerae	B	Cholera
	Hepatitis A virus	V	Infectious hepatitis
	Poliovirus	V	Poliomyelitis
	Trichinella spiralis	H	Trichinosis
Skin	*Clostridium tetani*	B	Tetanus
	Rickettsia rickettsii	B	Rocky Mountain spotted fever
	Rabies virus	V	Rabies
	Trichophyton rubrum	F	Tinea pedis (athlete's foot)
	Plasmodium vivax	P	Malaria
Genital tract	*Neisseria gonorrhoeae*	B	Gonorrhea
	Treponema pallidum	B	Syphilis
	Chlamydia trachomatis	B	Urethritis
	Human papilloma virus	V	Genital warts
	Candida albicans	F	Vaginitis

TABLE 7–7 Zoonotic Diseases Caused by Bacteria

Bacterium	Main Reservoir	Mode of Transmission	Disease
Gram-positive rods			
Bacillus anthracis	Domestic animals	Direct contact	Anthrax
Listeria monocytogenes	Domestic animals	Ingestion of unpasteurized milk products	Sepsis in neonate or mother
Erysipelothrix rhusiopathiae	Fish	Direct contact	Erysipeloid
Gram-negative rods			
Bartonella henselae	Cats	Skin scratch	Cat-scratch disease
Brucella species	Domestic animals	Ingestion of unpasteurized milk products; contact with animal tissues	Brucellosis
Campylobacter jejuni	Domestic animals	Ingestion of contaminated meat	Diarrhea
Escherichia coli O157:H7	Cattle	Fecal–oral	Hemorrhagic colitis
Francisella tularensis	Many animals, especially rabbits	Tick bite and direct contact	Tularemia
Pasteurella multocida	Cats	Cat bite	Cellulitis
Salmonella enteritidis	Poultry, eggs, and cattle	Fecal–oral	Diarrhea
Yersinia enterocolitica	Domestic animals	Fecal–oral	Diarrhea
Yersinia pestis	Rodents, especially rats and prairie dogs	Rat flea bite	Sepsis

- **Zoonotic disease:** **Transmission of a pathogen between animals (could be reservior host) and human being.**

You are expected to know all of these infectious agents by the third year of your studies.

TABLE 7–7 Zoonotic Diseases Caused by Bacteria

Mycobacteria			
Mycobacterium bovis	Cows	Ingestion of unpasteurized milk products	Intestinal tuberculosis
Spirochetes			
Borrelia burgdorferi	Mice	Tick bite (*Ixodes*)	Lyme disease
Leptospira interrogans	Rats and dogs	Urine	Leptospirosis
Chlamydiae			
Chlamydia psittaci	Psittacine birds	Inhalation of aerosols	Psittacosis
Rickettsiae			
Rickettsia rickettsii	Rats and dogs	Tick bite (*Dermacentor*)	Rocky Mountain spotted fever
Coxiella burnetii	Sheep	Inhalation of aerosols of amniotic fluid	Q fever
Ehrlichia chaffeensis	Dogs	Tick bite (*Dermacentor*)	Ehrlichiosis

You are expected to know all of these infectious agents by the third year of your studies.

2. Adherence to Cell Surfaces

Pathogens attach to host cells using specialized structures or substances called *adhesins:*

Major adhesins include:

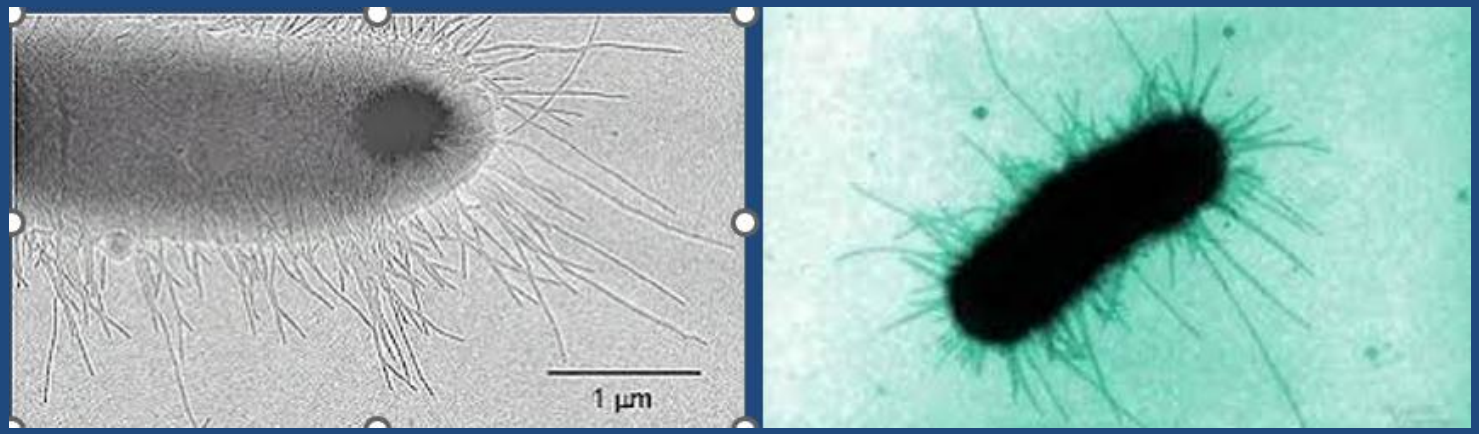
- **Pili (fimbriae)**
- **Capsules and glycocalyces**

Examples:

- **Pili of *Neisseria gonorrhoeae* and *Escherichia coli* mediate attachment to epithelial cells of the urinary and genital tracts.**

- **Glycocalyx of *Staphylococcus epidermidis* and certain viridans streptococci promotes strong adherence to damaged heart valves, facilitating colonization and biofilm formation.**

Pili (fimbriae)



Biofilms:

- Are structured communities of bacteria embedded in a matrix composed mainly of polysaccharides, proteins, and extracellular DNA, formed after bacterial attachment to a surface.

- Protect bacteria from antibiotics and host immune defenses, including antibodies and neutrophils.

- Contribute to chronic and persistent infections.

Examples:

Pseudomonas aeruginosa biofilms contribute to persistent infection in the lungs of cystic fibrosis patients.

Biofilms are an important component of dental plaque, which contributes to the development of dental caries.

Quorum Sensing: is a cell-to-cell communication system that allows bacteria to coordinate gene expression according to their population density.

- Bacteria produce and detect signaling molecules (autoinducers).

- When bacterial density reaches a critical threshold, signaling molecules accumulate and activate the expression of virulence factors, including factors involved in biofilm formation.

Example: *Pseudomonas aeruginosa* uses quorum sensing to regulate biofilm formation.

Bacterial Adhesion: Curli and Coagulation

Curli are extracellular amyloid fibers (fimbriae) produced by some strains of *E. coli* and *Salmonella*.

- They promote **bacterial adhesion** to:
  - -Endothelial cells
  - - Extracellular matrix proteins, such as fibronectin

Interaction with the Coagulation System: Curli can bind serum proteins, including factor XII and fibrinogen, that affecting coagulation systems, leading to fibrin formation and inflammatory responses.

Clinical Significance:

- **Coagulation disturbances** associated with severe bacterial infections and sepsis.
- Curli contribution to **thrombus formation and DIC**.

Disseminated intravascular coagulation (DIC)

DIC is systemic activation of coagulation in response to bacterial sepsis, resulting in widespread microvascular thrombosis, consumption of platelets and clotting factors, and potentially severe bleeding and organ dysfunction.



Examples of bacteria that can cause DIC:

- **Most Gram-negative bacteria that release endotoxin.**
- ***Neisseria meningitidis* (meningococcal sepsis).**
- ***Escherichia coli* (severe bloodstream infection).**
- ***Staphylococcus aureus* (severe sepsis).**
- ***Streptococcus pneumoniae* (invasive pneumococcal infection).**

Thrombus: is clotting process that prevents bleeding,
Thrombosis: when clot prevents blood flow in healthy blood vesseles.

3. Invasion, Inflammation, & Intracellular Survival

A- Invasion.

Bacterial enzymes and toxins promote tissue invasion, spread, and evasion of host immune defenses.

1. Collagenase and hyaluronidase: degrade collagen hyaluronic acid in the extracellular matrix, facilitating bacterial spread through tissues.

- Example: spread of *Streptococcus pyogenes* in skin and soft-tissue infections, including cellulitis.

2. Coagulase: Produced by *Staphylococcus aureus* and associated with *Yersinia pestis*.

Promotes fibrin clot formation from fibrinogen, potentially protecting bacteria from phagocytosis by surrounding organisms or walling off the infected area.

3. Invasion, Inflammation, & Intracellular Survival

3. Immunoglobulin proteases:

- **IgA protease:** Degrades secretory IgA on mucosal surfaces. Produced by *Neisseria gonorrhoeae*, *Haemophilus influenzae*, and *Streptococcus pneumoniae*.
- **IgG-degrading protease:** Produced by *Streptococcus pyogenes*.

4. Leukocidins

- Cytotoxins that damage or kill leukocytes, particularly neutrophils and macrophages.
- Many leukocidins form pores in host-cell membranes.

Cellulitis: Bacterial skin infection that affects the deeper layers of the skin. It typically occurs when bacteria enter the body through a break, cut, scratch, or crack in the skin

- Appears as a swollen, red area of skin
- Caused by *Streptococcus pyogenes.*





Antiphagocytic Virulence Factors

- Virulence factors that inhibit opsonization and phagocytosis.

1. Polysaccharide Capsule

Examples: *Streptococcus pneumoniae* and *Neisseria meningitidis*.

The capsule inhibits phagocytosis by interfering with the deposition and accessibility of opsonins, such as C3b and IgG, on the bacterial surface.

Opsonization enhances phagocytosis: C3b and IgG coat the bacterial surface and facilitate recognition by complement receptors and Fc receptors on phagocytes.

Antiphagocytic Virulence Factors

2. Antiphagocytic Cell-Surface Proteins

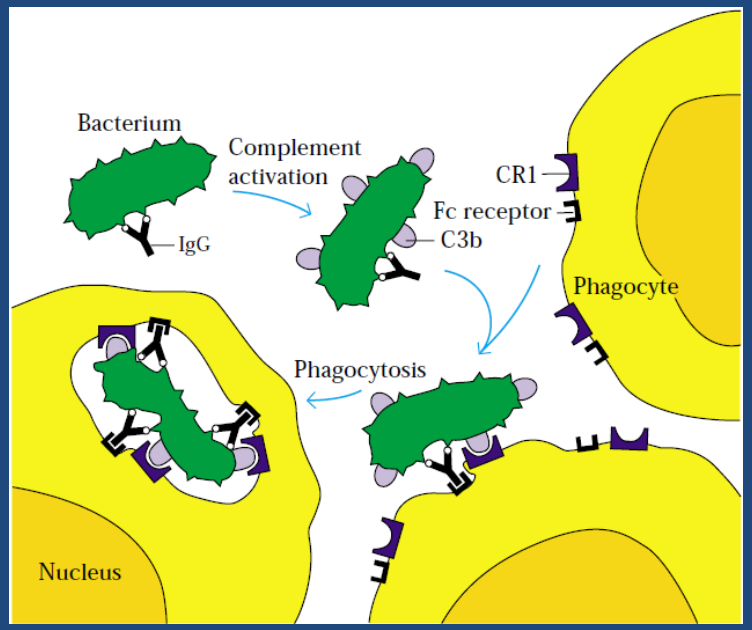
A. M protein, as in *Streptococcus pyogenes* (Group A Streptococcus)

- M protein binds factor H, a host complement-regulatory protein, which promotes the inactivation of C3b.

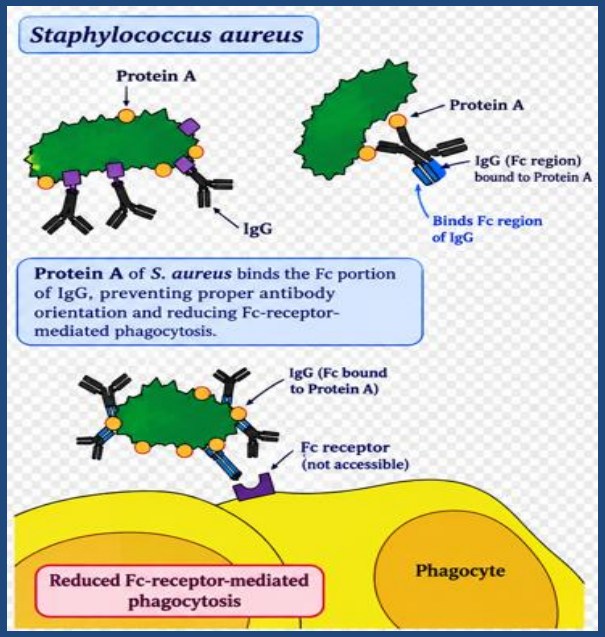
B. Protein A, as in *Staphylococcus aureus*

- Protein A binds the Fc region of IgG, interfering with the normal orientation of antibodies on the bacterial surface.

Opsonization and Phagocytosis of Bacteria by Phagocytes



Phagocytosis without the action of antiphagocytic virulence factors



Antiphagocytic virulence factors

- IgG antibodies bind to the bacterial surface.
- The Fc receptors on phagocytes recognize the Fc portion of IgG.
- Complement receptor 1 (CR1) binds C3b.
    - These interactions enhance bacterial recognition, attachment, and subsequent phagocytosis.

TABLE 7–8 Surface Virulence Factors Important for Bacterial Pathogenesis

Organism	Virulence Factor	Used in Vaccine	Comments
Gram-positive cocci			
Streptococcus pneumoniae	Polysaccharide capsule	Yes	Determines serotype
Streptococcus pyogenes	M protein	No	Determines serotype[1]
Staphylococcus aureus	Protein A	No	Binds to Fc region of IgG
			Prevents activation of complement
Gram-negative cocci			
Neisseria meningitides	Polysaccharide capsule	Yes	Determines serotype
Gram-positive rods			
Bacillus anthracis	Polypeptide capsule	No	
Gram-negative rods			
Haemophilus influenzae	Polysaccharide capsule	Yes	Determines serotype
Klebsiella pneumoniae	Polysaccharide capsule	No	
Escherichia coli	Protein pili	No	Causes adherence
Salmonella typhi	Polysaccharide capsule	No	Not important for other
Yersinia pestis	V and W proteins	No	

3. Invasion, Inflammation, & Intracellular Survival

B- Inflammation: Bacteria can cause two types of inflammation

1- Pyogenic (pus-producing / Suppurative) inflammation, neutrophils are the predominant cells. most important pyogenic bacteria are some gram-positive and gram-negative cocci

2- Granulomatous inflammation: macrophages and helper T cells predominate. The most important organism: *Mycobacterium tuberculosis.*

Phagocytosis by macrophages kills most of the bacteria, but some survive and grow within the macrophages in the granuloma.

Chronic Granulomatous Disease (CGD)

- CGD is a group of inherited primary immunodeficiency disorders caused by defects in the phagocyte NADPH oxidase complex.

- This defecte affects the respiratory burst and the production of reactive oxygen species (ROS).

- Consequently, phagocytes, (neutrophils and macrophages), have impaired intracellular killing of certain bacteria and fungi.

- Persistent infections and dysregulated inflammation can lead to granuloma formation in various organs, sometimes causing tissue damage or obstruction.

Another Pattern of Inflammation: Pseudomembrane Formation

Pseudomembranes are inflammatory exudates that form on damaged mucosal surfaces. **Two Important Examples:**

1. Diphtheria: caused by *Corynebacterium diphtheriae* (toxigenic strains).

- A thick, grayish-white, firmly adherent pseudomembrane forms in the pharynx or tonsils.
- The diphtheria toxin causes cellular injury, leading to an exudate containing inflammatory cells, dead epithelial cells, and bacteria.

2. Pseudomembranous Colitis: caused by *Clostridioides difficile*. Often develops following disruption of the normal intestinal microbiota, particularly after antibiotic use.

3. Invasion, Inflammation, & Intracellular Survival

C- Intracellular Survival of Pathogens

Intracellular survival enables certain pathogens to **evade host** immune defenses. Some persistent intracellular pathogens induce **granulomatous inflammation**

- Facultative intracellular pathogens: Example: *Mycobacterium, Legionella, Brucella,* and *Listeria. Histoplasma (fungi).*

- Obligate intracellular pathogens: Require host cells for replication. Examples include *Chlamydia* spp. and *Rickettsia* spp.

Bacterial mechanisms for intracellular survival:

1- Inhibition of the fusion of the phagosome with the lysosome (*Mycobacterium and Legionella).*

2- Inhibition of acidification of the phagosome, which reduces the activity of the lysosomal degradative enzymes (*Mycobacterium and Legionella)*

3- Escape from the phagosome into the cytoplasm, where there are no degradative enzymes. (as in *Listeria)*

Bacterial Invasins and Host Cell Invasion

Invasins are bacterial surface proteins that promote bacterial entry into host cells by interacting with specific host-cell receptors.

Some invasins bind to integrins, transmembrane cell-adhesion receptors, triggering intracellular signaling that promotes bacterial internalization.

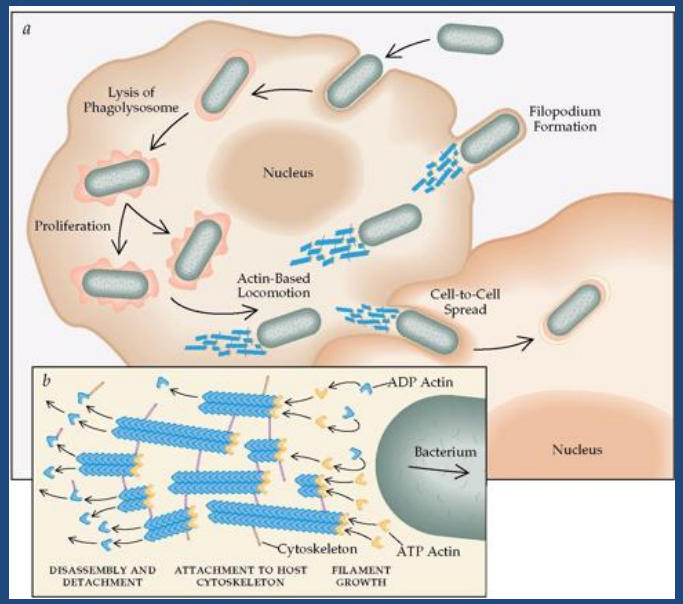
Once inside the cell:

1 - Some bacterial cells **reside within cell vacuoles** (phagosomes).

2 -Others; **migrate into the cytoplasm.**

3 - Some move from the cytoplasm **into adjacent cells** through tunnels formed from **actin**.

Actin-Based Motility and Cell-to-Cell Spread of Intracellular Bacteria
Example: *Listeria monocytogenes*



Infection of the surrounding cells: in this manner the bacteria can evade host defenses. For example, *Listeria monocytogenes* aggregates actin filaments on its surface and is propelled in a "sling-shot" fashion, called actin rockets, from one host cell to another.

Yersinia Outer Proteins (Yops): Mechanisms of Bacterial Immune Evasion

Yersinia outer proteins (Yops) are virulence factors produced *Yersinia* species, including ***Yersinia pestis***, the causative agent of **plague.**

There are about 12 different Yop effector proteins.

Major Effects of Yops

1- Inhibition of phagocytosis: Prevent neutrophils and macrophages from engulfing and eliminating bacteria.

2- Suppression of cytokine production: Reduce inflammatory responses, including tumor necrosis factor (TNF) production by macrophages.

Example: YopJ is a protease that cleaves signal transduction proteins required for the induction of TNF synthesis.

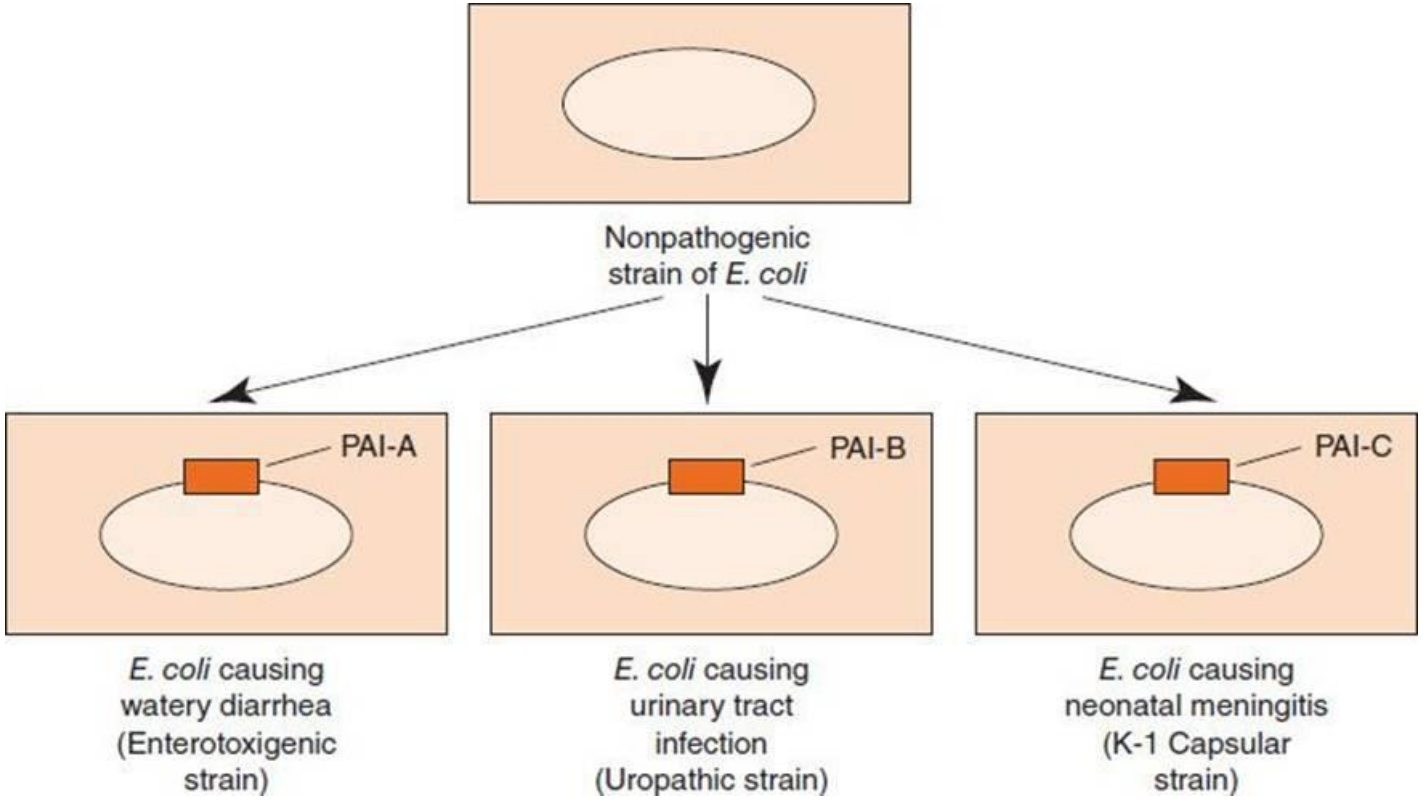
Pathogenicity Islands: Genetic Basis of Bacterial Virulence

- Pathogenicity islands (PAIs): Clusters of genes encoding multiple virulence factors on the bacterial chromosome.

- Genes encoding adhesins, invasins, and exotoxins are located adjacent to each other on these islands.

- Nonpathogenic variants of these bacteria lack pathogenicity islands. These large genomic regions appear to have been transferred as a block via conjugation or transduction.

- Pathogenicity islands are found in many Gram-negative rods, such as *E. coli*, *Salmonella*, *Shigella*, *Pseudomonas*, and *Vibrio cholerae*, and Gram-positive cocci, such as *S. pneumoniae*.



Pathogenicity islands encode virulence factors that determine the type of infection. The top of the figure depicts a nonpathogenic strain of *Escherichia coli that does not contain a pathogenicity island (PAI) in the genome DNA.*

PAI-A encodes an enterotoxin.
PAI-B encodes the pili that bind to urinary tract epithelium.
PAI-C encodes the enzymes that synthesize the K-1 capsular polysaccharide.